A Thought Experiment

Suppose you had a container full of red liquid. You add blue liquid into it slowly, mixing as you go. At the same time, you drain off well-mixed liquid at the same rate you're adding red liquid. What do you expect to happen to the color of the liquid over time? Explain.

Mixing Problems

Container problems are another class of situations we can model using the kind of differential equations we've been studying. The idea is this: if we have a container into which we add a substance at a certain rate and remove the substance at another rate, then the rate of change of the amount of the substance in the container is given by the difference between the two rates. That is:

$$\text{rate of change} = \text{rate in} - \text{rate out}$$

1. □Regrade An 800 gallon tank is filled with brine which contains 300 pounds of salt. Every minute two gallons of dilute brine containing 0.1 pound of salt per gallon are pumped in and two gallons of the (well-mixed) brine are pumped out. Find a differential equation and initial condition satisfied by $S(t)$, the weight (in lbs) of salt in the tank at time t. Solve the initial value problem to find $S(t)$. Draw a graph of $S(t)$ and describe its behavior as $t \to \infty$. Explain why your answer makes sense.

2. □Regrade A tank of salt water has 100 pounds of salt dissolved in 1000 gallons of water. Every minute four gallons of salt water having 0.5 pounds of salt per gallon are pumped in and four gallons of the (well-mixed) salt water are pumped out. Find a differential equation and initial condition satisfied by $S(t)$, the weight (in lbs) of salt in the tank at time t. Solve the initial value problem to find $S(t)$. Draw a graph of $S(t)$ and describe its behavior as $t \to \infty$.

Examining the Differential Equation

In both examples above, we were adding a fixed amount of liquid per minute to a container, and removing the same amount per minute. That is, the total _____ of liquid never changed. We will consider what happens when it does change on a later worksheet.

For now, let's stick to that assumption and see if we can understand the equation in a similar way to how we understood k in the context of cooling.

Suppose that:

- The volume of liquid is V.
- We begin with S_0 lbs of salt dissolved in the liquid.
- The liquid we're adding has concentration C lbs of salt per gallon.
- We're adding it at a rate of R gallons per minute, and we're draining the well-mixed tank at the same rate.

Let $S(t)$ be the amount of salt in the tank at time t.

In problem 2 above, what are the values of V, S_0, C, and R? Include units.

3. □Regrade (a) Explain why we expect the amount of salt in the tank to approach VC in the long term.

(b) What does the quantity CR represent? Explain. What are its units?

4. ☐Regrade (a) Suppose that we increase R, the rate at which we're adding in liquid (and draining it). Do you expect that we approach the eventual final concentration faster or slower? Explain.

(Hint: think back to the thought experiment with the colors. If you add blue stuff at a faster rate, what do you expect to happen?)

(b) Instead, suppose we don't change the rate of adding/draining, but instead think about a far larger container. Do you expect that we approach the eventual final concentration faster or slower? Explain.

(Hint: think colors again.)

5. ☐Regrade (a) If the amount of salt in the container at time t is $S(t)$ lbs, and the total volume of liquid is V gal, then the concentration at time t is ____ _____ (second blank: units).

(b) We are removing R gal/min of this liquid. How many lbs of salt are being removed each minute?

6. ☐Regrade Explain why

$$\frac{dS}{dt} = CR - \frac{R}{V}S.$$

7. In problems 1 and 2 above, we factored out the coefficient of S before separating ☐Regrade variables to solve. Here, the coefficient of S is $-\frac{R}{V}$. Factor that out of the differential equation in the previous question to show that

$$\frac{dS}{dt} = -\frac{R}{V}\left(-VC + S\right).$$

What is the equilibrium value? Is it stable or unstable? Compare Question 3(a) and explain.

8. Newton's law of cooling was $\frac{dT}{dt} = -k(T - T^*)$. We found that the larger k was, the ☐Regrade _____ the object's temperature approached the ambient temperature.

(a) We saw that VC plays the same role here as T^* did there. What is the analogue here of k?

(b) Explain how this lines up with Question 4.